RESEARCH ARTICLE | SOFTWARE AND POPULATION GENETICS

An Auditable qpAdm Workflow for South Asian Population-Genetic Modelling

A repository evaluation and literature-derived benchmark of the Indian Cline

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Manuscript scope

This paper evaluates a notebook-oriented workflow that prepares genotype data, interfaces with classic ADMIXTOOLS qpAdm, and organizes analytical outputs. It also presents a literature-derived Indian Cline benchmark for interpretation. It does not introduce a new ancestry statistic, and it does not claim that all proposed reproducibility controls have already been validated.

Project repository

github.com/AELIUS-22/qpAdm-notebook

Portfolio research document. Not a peer-reviewed publication.

Abstract

South Asian population history is often discussed through clinal variation involving ancestry related to a deeply divergent South Asian lineage, populations on the Indus Periphery Cline, and Bronze Age Steppe groups. Such labels are useful only when their proxy status and inferential limits remain explicit. This paper has two connected aims. First, it audits a notebook-oriented qpAdm project that links consumer-format genotype preparation, PLINK conversion, EIGENSTRAT preparation, AADR merging, classic ADMIXTOOLS execution, parsing, and descriptive plotting. Second, it organizes a population-level benchmark compiled from published qpAdm results for selected South Asian samples. The benchmark is retained as literature-derived evidence and is not presented as newly generated output from the repository. The software contribution is framed narrowly: integration, workflow documentation, provenance, and an evaluation design around an established statistical engine. The audit identifies preparation and single-model execution components present in the project while distinguishing them from controls that still require end-to-end validation, including complete build/allele/strand harmonization, general batch scheduling, robust failure capture, repeat-run testing, and matched comparison with direct ADMIXTOOLS. Published coefficients show broad differences among the selected benchmark rows, but rejected models, small samples, missing standard errors, and population-specific structure prevent simple regional generalization. The paper concludes with a practical validation roadmap and reporting rules intended to improve transparency without overstating biological or software conclusions.

Keywords: qpAdm; ADMIXTOOLS; AADR; South Asia; Indian Cline; genotype harmonization; population structure; reproducibility; auditability; literature benchmark.

1. Introduction

Ancient-DNA research has refined models of South Asian population history and encouraged the use of formal statistical frameworks to evaluate mixture hypotheses. A simplified three-component description is sometimes used to communicate broad gradients: AASI-related ancestry, Indus Periphery-related ancestry, and Steppe_MLBA-related ancestry. These terms describe affinities to selected reference populations or lineages. They should not be treated as literal, fully sampled ancestral populations, fixed identities, or direct proxies for language and culture [1, 13, 21-23].

A qpAdm result is conditional on the target, proposed sources, right/reference populations, SNP set, data processing, and software configuration. The right set is part of the tested hypothesis. A model that is not rejected at a prespecified significance threshold is not thereby proven to be the unique or historically correct explanation. More than one model may be statistically compatible with the same data, especially where source proxies are imperfect or demographic histories are complex [2-5, 8].

This study treats the computational workflow and the biological interpretation as linked but distinct tasks. A reliable interface should make each transformation and model choice inspectable. At the same time, a well-organized pipeline cannot compensate for weak source proxies, incompatible genotypes, inadequate reference panels, or non-identifiability in the underlying demographic history.

2. Aims and contribution

The paper addresses four questions: (i) which preparation and analysis stages are evidenced in the current project; (ii) which controls are still needed for a defensible reproducibility claim; (iii) how published South Asian studies constrain the interpretation of the Indian Cline benchmark; and (iv) what can be concluded from the available literature-derived coefficients before new, matched project runs are completed.

The contribution is an evidence-aware software and research audit, not a new statistical method. It documents an integration layer around classic ADMIXTOOLS, preserves a literature benchmark with model-fit status, and specifies tests needed to compare the wrapper with direct engine execution.

3. Statistical and data foundations

3.1 qpAdm and model interpretation

qpAdm uses f-statistic relationships to evaluate whether a target population is compatible with a mixture of specified source populations relative to a selected right set [2, 3]. The test is not a general-purpose ancestry percentage calculator. Coefficients are conditional on the model and input data; p-values address model compatibility under the tested configuration. At alpha = 0.05, $p > 0.05$ should be reported as “not rejected,” while $p < 0.05$ indicates rejection of that specified model under the test. Neither outcome independently establishes a complete population history.

3.2 AADR and genotype compatibility

The Allen Ancient DNA Resource (AADR) is a curated, versioned resource for ancient and present-day genotype data [6]. The 1240K panel is widely used in ancient-DNA research [7]. However, matching marker identifiers is only an initial overlap screen. A defensible merge must also consider genome build, chromosome and position, allele coding, strand orientation, ambiguous A/T and C/G sites, duplicate markers, missing calls, and the number of sites retained at each stage. The final model-usable SNP count should be reported rather than inferred from an early filtering count.

3.3 Existing software context

Classic ADMIXTOOLS provides the statistical engine and associated conversion/merge utilities. ADMIXTOOLS 2 offers an R-based implementation and multi-model functions; qpAdm-wrapper and ColabADMIXTOOLS provide prior workflow precedents [9-12]. PLINK is an established tool for genotype-data manipulation [20]. These projects establish that qpAdm execution and repeated-model workflows are not novel by themselves. The present repository should therefore be evaluated on the quality of its preparation, configuration, logging, provenance, and validation rather than on a broad first-of-its-kind claim.

System	Relevant capability	Boundary for this project
Classic ADMIXTOOLS	Classic qpAdm, qpfstats, convertf and mergeit utilities.	The wrapper should preserve engine versions, parameters and raw outputs.
ADMIXTOOLS 2	R-based f-statistics and multi-model/rotation functionality.	Do not attribute established engine capabilities to the wrapper itself.
qpAdm-wrapper / ColabADMIXTOOLS	Prior examples of repeated qpAdm runs and notebook workflows.	These are relevant prior art; the project must define its narrower contribution.
Current qpAdm-notebook	Preparation steps, classic single-model execution, parsing and plotting are represented in the audited project materials.	Complete harmonization, general batch ledger, repeatability and deployment audit remain pending.

Table 1. Software context and the appropriate scope of the project contribution.

4. Population-genetic framework and evidence base

4.1 Interpreting the three analytical components

AASI-related. This label refers to affinity with a deeply divergent South Asian lineage that is not directly represented by a fully sampled ancient genome in the benchmark model. Onge or other Andamanese-related groups are imperfect proxies and should not be described as unchanged or literal AASI ancestors.

Indus Periphery-related. This is a model-dependent label for ancestry related to populations on the Indus Periphery Cline. Such populations themselves reflect ancestry from more than one source; the label should not be collapsed into a simple “Iranian farmer” identity.

Steppe_MLBA-related. This refers to affinity with Middle-to-Late Bronze Age Steppe populations used as proxies in the model. It is a genetic reference label, not a direct measure of language, ethnicity, or cultural identity.

4.2 Why community-level structure matters

South Asian studies report substantial variation across and within regions. Endogamy, founder effects, migration, sex-biased gene flow, and local demographic history can create structure that is obscured by broad state labels. Research on Kumhar and Kurcha groups, West Coast Roman Catholic populations, South-West Coast populations, Coorg communities, and Bihar caste groups illustrates the importance of sample definition and study design [14-19]. Genetic and oral histories also occupy distinct evidential domains and should be interpreted together rather than treated as interchangeable [18].

The benchmark therefore retains population or panel labels and does not treat a state as a homogeneous biological unit. Approximate sample counts are reproduced as reported in the source manuscript; “NR” indicates that the value was not independently verified in the accessible benchmark material.

5. Data provenance and literature benchmark

The coefficients below were transcribed from the source manuscript's consolidated literature benchmark. They are not new qpAdm runs from the GitHub repository. Percentages are rounded to one decimal place. Standard errors are unavailable in the accessible benchmark and are therefore not supplied. The Onge/AHG proxy row is a reference-source description and is not included as a modeled target.

ID	Region	Population / panel	N	AASI-rel. (%)	Indus Per. (%)	Steppe (%)	p-value	Status
ULL-KL	Kerala	Ulladan	6	61.9	35.7	2.4	0.2386	Not rejected
PAL-TN	Tamil Nadu	Palliyar	11	64.7	33.0	2.3	0.0337	Rejected
IRU-TN	Tamil Nadu	Irula	NR	57.3	40.1	2.5	0.2489	Not rejected
LOH-GJ	Gujarat	Lohana	2	10.4	64.3	25.2	0.0517	Not rejected
GUJA-GJ	Gujarat ancestry / diaspora	GujaratiA panel	4	11.6	63.9	24.5	0.3449	Not rejected
YAD-RJ	Rajasthan	Yadav_Rajasthan	3	15.8	61.6	22.6	0.0765	Not rejected
BRH-HR	Haryana	Brahmin_Haryana	2	18.2	57.2	24.6	0.4269	Not rejected
RAJ-HR	Haryana	Rajput	3	20.3	58.4	21.2	0.0177	Rejected
BHU-BR	Bihar	Bhumihar_Bihar	7	20.7	52.7	26.6	0.0514	Not rejected
YAD-UP	Uttar Pradesh	Yadav_UP	3	17.3	61.4	21.3	0.2081	Not rejected
BHU-UP	Uttar Pradesh	Bhumihar_UP	8	23.4	53.2	23.4	0.0174	Rejected
BRH-UP	Uttar Pradesh	Brahmin_UP	7	24.1	50.2	25.7	0.2591	Not rejected

Table 2. Consolidated population-level literature benchmark. “Not rejected” and “Rejected” refer only to the specified published model and its reported qpAdm p-value.

Interpretive caution. The benchmark includes small sample counts, a diaspora/study panel, unavailable standard errors, and rejected models. It must not be presented as a representative survey of all residents of the named states or communities. Coefficients from rejected models are retained for transparency but should not be promoted as accepted ancestry reconstructions.

6. Software workflow and repository audit

6.1 Workflow represented in the project

The project materials describe a workflow that begins with consumer-format genotype input and proceeds through PLINK conversion, EIGENSTRAT preparation using convertf, merging with AADR using mergeit, SNP-count reporting, model specification, classic qpAdm execution, parsing, and descriptive plotting. The statistical engine remains ADMIXTOOLS; the repository contribution is the surrounding preparation and organization of the analysis.

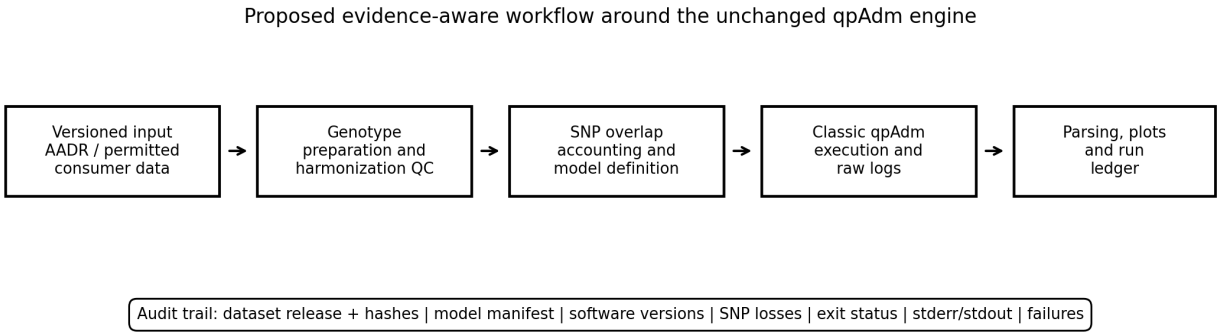


Figure 1. Evidence-aware workflow concept. The audit trail is a design requirement; not every control shown has been demonstrated end-to-end in the current repository.

6.2 What the repository audit supports

The supplied repository audit identifies PLINK-based consumer-format conversion, convertf/mergeit preparation steps, qpfstats and classic qpAdm invocation, and parsing/plotting of a saved result. It also identifies a single-model interaction pattern and limitations in robust status handling. These observations support a careful description of an implemented workflow prototype, not a claim of a fully validated production pipeline.

Stage	Evidence described in audit	Outstanding validation
Input / SNP filtering	Filtering logic and optional marker extraction are present.	Report build, position, allele orientation, ambiguous sites, duplicates and per-stage losses.
Consumer import	PLINK --23file conversion is present.	Document supported vendor formats and validate parser assumptions with fixtures.
AADR preparation	convertf/mergeit workflow is present; allele warnings were noted in stored output.	Resolve and report harmonization issues; record usable SNP counts after merge and per model.
Model definition	Notebook accepts target and source/right population lists.	Freeze a versioned machine-readable manifest and distinguish benchmark, sensitivity and exploratory runs.
Statistics	qpfstats and classic qpAdm calls are represented.	Capture return codes, stdout/stderr, runtime and failed attempts.
Parsing / plotting	Saved output can be parsed into coefficients and fit fields and plotted.	Test parser fixtures across output versions; retain rejected and malformed runs.
Batch / deployment	General batch execution and deployed application provenance were not demonstrated in the supplied audit.	Audit source, dependencies and deployment; claim batch or PCA only after tests and evidence exist.

Table 3. Repository audit: observed capabilities and required evidence.

6.3 Reproducibility requirements

A reproducible run should be tied to a machine-readable manifest containing the target, ordered sources, right populations, SNP handling rule, AADR release, input hashes, ADMIXTOOLS version, parameter file, command, and output paths. A run ledger should preserve every attempt, including failures and rejected models, with exit status, standard output, standard error, runtime, SNP counts, coefficients, standard errors, p-value, and status. Deterministic run identifiers are useful only when their inputs and hash procedure are documented.

6.4 Matched comparison with direct ADMIXTOOLS

The primary software evaluation should run the same target, sources, right set, genotype files, parameter files, and engine version directly and through the wrapper. Compare printed coefficients, standard errors, fit statistics, usable SNP counts, and failure behavior. Secondary measures can include elapsed time, manual steps, and ledger completeness. At least one intentionally invalid input should test whether errors are retained rather than silently discarded.

6.5 Keep formal qpAdm and coordinate-space analyses separate

Global25/Vahaduo-style coordinate fits can be useful exploratory visualizations, but their distances are not equivalent to qpAdm rank-test p-values. They should be reported in a separate evidence track and must not be combined into a shared “model validity” ranking.

7. Benchmark visualization and descriptive results

The stacked chart below visualizes the same literature-derived coefficients shown in Table 2. It is descriptive: it does not override model-fit status, recover missing uncertainty, or establish a unique demographic history. Southern rows in this selected panel have higher AASI-related coefficients and low Steppe_MLBA-related coefficients; the selected western and northern rows show larger Indus Periphery-related and Steppe_MLBA-related coefficients. These patterns describe the included rows only.

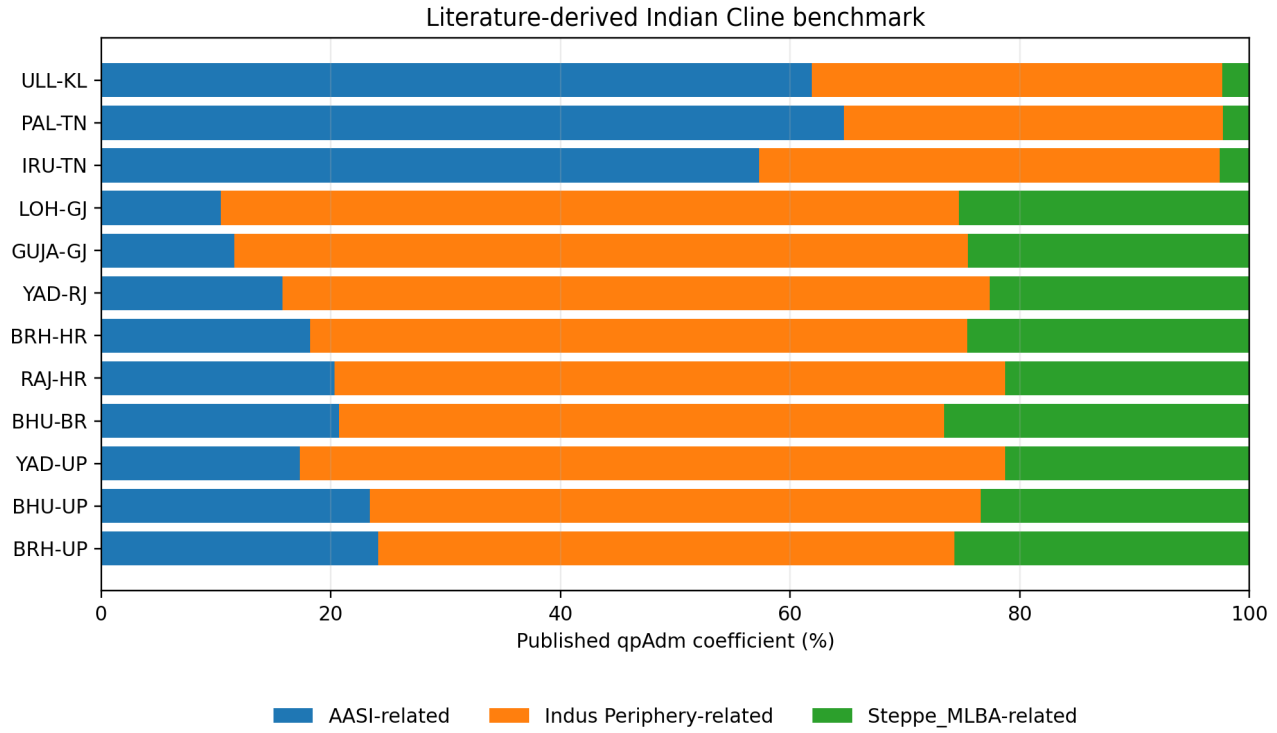


Figure 2. Stacked coefficients for the selected published benchmark rows. The plot is not a state-level estimate and includes rejected models for transparent display.

The source manuscript reports unweighted descriptive means of approximately 61.3% AASI-related, 36.3% Indus Periphery-related, and 2.4% Steppe_MLBA-related for the southern subset; 12.6%, 63.3%, and 24.1% for Gujarat/Rajasthan; and 20.7%, 55.5%, and 23.8% for the Haryana/Bihar/Uttar Pradesh subset. These are arithmetic summaries of the displayed rows, not population-weighted estimates or inferential group comparisons.

Formal ANOVA or chi-square testing is not justified from these rounded coefficients as if they were independent raw observations with uniform uncertainty. Sample sizes differ, some standard errors are unavailable, and some models are rejected. Future group-level inference requires a harmonized analysis dataset, consistent model definitions, and appropriate uncertainty estimates.

8. Y-chromosome and uniparental evidence boundary

Autosomal qpAdm coefficients and Y-chromosome haplogroups describe different inheritance pathways. A Y-chromosome call follows one paternal lineage and cannot be reconstructed from an aggregate autosomal benchmark row. The source benchmark therefore does not assign terminal Y haplogroups to these populations. Individual-level calls require suitable Y-SNP data, a documented caller and reference tree, marker coverage, and a stated resolution. Population frequency claims require appropriately sampled individuals and cannot be inferred from isolated cases.

9. Limitations and research-integrity safeguards

The present paper has several important limits. First, its ancestry coefficients are literature-derived and do not represent new project-generated results. Second, the AASI-related component is proxy-dependent; no directly sampled AASI genome is represented in the model. Third, sample sizes are small or unavailable for several rows, and some labels refer to panels or diaspora samples rather than collection sites. Fourth, model non-rejection is not proof of a unique history, while rejected models should not be interpreted as accepted reconstructions. Fifth, state names are descriptive geography and should not be used to generalize to all residents. Sixth, the current audit does not establish complete allele/strand harmonization, general batch execution, repeat-run reproducibility, or an independently audited deployment. Finally, even perfect software bookkeeping cannot resolve weak source proxies, inadequate right-set discrimination, unmodelled drift, or demographic non-identifiability.

Genetic ancestry must not be used to rank communities or make claims about cultural legitimacy. Oral, archaeological, linguistic, and genetic evidence address different questions. Any use of identifiable consumer genotype data requires explicit permission, data minimization, secure handling, and a clear retention/deletion policy.

10. Validation roadmap

Phase	Required work	Evidence to publish
1. Provenance freeze	Select and record one archived AADR release, panel definition, software versions and source publications.	Release identifiers, checksums, exact labels and environment record.
2. Harmonization QC	Implement build/position/allele/strand checks and deterministic SNP-loss accounting; add synthetic/public fixtures.	Stage-by-stage counts, reason-coded losses and QC report.
3. Manifest and ledger	Represent models in machine-readable form; capture run IDs, commands, return codes, logs and failures.	Frozen manifests and complete run ledger.
4. Matched benchmark	Run identical models directly and through the wrapper with the same engine and inputs.	Numerical fidelity report and workflow-cost comparison.
5. Repeatability	Repeat identical runs and test malformed, missing and low-overlap inputs.	Regression fixtures, output deltas and retained failure cases.
6. Population application	Only after earlier phases pass, run predeclared case studies and sensitivity analyses.	Complete model ledger, QC report and cautious interpretation.
7. Optional extensions	Audit any deployed app and add exploratory coordinate/uniparental modules only when justified.	Deployment provenance and separate exploratory outputs.

Table 4. Proposed sequence for completing the reproducibility evaluation.

11. Discussion

The project has two distinct forms of value. The software work can reduce manual friction and make genotype preparation and qpAdm execution easier to inspect. The biological benchmark provides a carefully qualified reference for discussing the broad Indian Cline pattern in selected published samples. Neither layer should be allowed to overstate the other: a polished interface does not validate a demographic model, and a published benchmark does not prove that the repository independently reproduced those estimates.

The most important software safeguard is preservation of the full analytical path. Inputs, transformations, source/right choices, SNP losses, engine versions, raw logs, failures, rejected models, and parser outputs should remain linked. The most important biological safeguard is conditional interpretation: every coefficient belongs to a specified model and reference configuration. Multiple compatible models may remain, and independent archaeological, chronological, linguistic, and population-genetic evidence is needed to evaluate historical explanations.

12. Conclusion

This revised manuscript presents the qpAdm-notebook project as an auditable workflow prototype around established ADMIXTOOLS functionality and places its South Asian interpretation within a literature-derived Indian Cline benchmark. The evidence supports a focused claim: the project materials describe genotype preparation, classic qpAdm invocation, result parsing and plotting, while several controls needed for full reproducibility remain to be implemented or demonstrated. The benchmark illustrates population-specific variation but is constrained by proxy choice, sample size, unavailable uncertainty, and model-fit status.

The next empirical milestone is a frozen, public-data end-to-end evaluation. It should record the exact AADR release, harmonization losses, model manifest, raw logs, fit outputs, failures, and repeat-run agreement, followed by a matched direct-ADMIXTOOLS comparison. Only after those tests pass should the paper report new project-generated population analyses or stronger reproducibility claims.

Data and code availability

Project repository: https://github.com/AELIUS-22/qpAdm-notebook. The repository should identify the exact commit and dependency environment used for any future empirical result. AADR is distributed through its official data access channels [6]. This manuscript does not redistribute personal consumer-genotype files.

Ethics statement

This manuscript synthesizes published literature, public software documentation, and a repository audit. It does not report recruitment of new human participants or new individual-level consumer-genotype analysis. Future use of identifiable genotype data requires explicit informed permission and a documented data-protection procedure.

Competing interests

The project team reports no competing interests for this manuscript draft.

Author note

Author credit retained from the source manuscript: qpAdm Wrapper Project Team. Institutional affiliation and individual contributor roles were not expanded because no verified details were supplied in the source document.

References

1. Narasimhan VM, Patterson N, Moorjani P, et al. The formation of human populations in South and Central Asia. Science. 2019;365(6457):eaat7487. doi:10.1126/science.aat7487.

2. Patterson N, Moorjani P, Luo Y, et al. Ancient admixture in human history. Genetics. 2012;192:1065-1093. doi:10.1534/genetics.112.145037.

3. Harney E, Patterson N, Reich D, Wakeley J. Assessing the performance of qpAdm: a statistical tool for studying population admixture. Genetics. 2021;217(4):iyaa045. doi:10.1093/genetics/iyaa045.

4. Maier R, Flegontov P, Flegontova O, Isildak U, Changmai P, Reich D. On the limits of fitting complex models of population history to f-statistics. eLife. 2023;12:e85492. doi:10.7554/eLife.85492.

5. Flegontova O, Isildak U, Yuncu E, et al. Performance of qpAdm-based screens for genetic admixture on graph-shaped histories and stepping stone landscapes. Genetics. 2025;230:iyaf047. doi:10.1093/genetics/iyaf047.

6. Mallick S, Micco A, Mah M, et al. The Allen Ancient DNA Resource (AADR): a curated compendium of ancient human genomes. Scientific Data. 2024;11:182. doi:10.1038/s41597-024-03031-7.

7. Rohland N, Mallick S, Mah M, et al. Three assays for in-solution enrichment of ancient human DNA at more than a million SNPs. Genome Research. 2022;32:2068-2078. doi:10.1101/gr.276728.122.

8. Williams MP, Flegontov P, Maier R, Huber CD. Testing times: disentangling admixture histories in recent and complex demographies using ancient DNA. Genetics. 2024;228:iyae110. doi:10.1093/genetics/iyae110.

9. ADMIXTOOLS 2 documentation. qpadm, qpadm_multi and qpadm_rotate. Accessed October 2026.

10. David Reich Lab. ADMIXTOOLS source repository. Accessed October 2026.

11. Koptekin D. qpAdm-wrapper repository and documentation. Accessed October 2026.

12. ColabADMIXTOOLS repository and documentation. Accessed October 2026.

13. qpAdm-notebook project repository. Repository audit supplied for commit 2d3641a4b29e64d3e06095ab53a2bed8f2526d4f; public README cross-checked October 2026.

14. Das R, Ivanisenko VA, Anashkina AA, Upadhyai P. The story of the lost twins: decoding the genetic identities of the Kumhar and Kurcha populations from the Indian subcontinent. BMC Genetics. 2020;21(Suppl 1):117. doi:10.1186/s12863-020-00919-2.

15. Kumar L, Farias K, Prakash S, et al. Dissecting the genetic history of the Roman Catholic populations of West Coast India. Human Genetics. 2021;140(10):1487-1498. doi:10.1007/s00439-021-02346-4.

16. Kumar L, Chowdhari A, Sequeira JJ, Mustak MS, Banerjee M, Thangaraj K. Genetic affinities and adaptation of the South-West Coast populations of India. Genome Biology and Evolution. 2023;15(12):evad225. doi:10.1093/gbe/evad225.

17. Mukhopadhyay A, Kumar L, Sran K, Thangaraj K, Thelma BK. Unique demographic history and population substructure among the Coorgs of Southern India. Communications Biology. 2025;8:698. doi:10.1038/s42003-025-08073-0.

18. Biddanda A, Bandyopadhyay E, de la Fuente Castro C, et al. Distinct positions of genetic and oral histories: Perspectives from India. Human Genetics and Genomics Advances. 2024;5(3):100305. doi:10.1016/j.xhgg.2024.100305.

19. Ashma R, Kashyap VK. Genetic profile based upon 15 microsatellites of four caste groups of the eastern Indian state, Bihar. Annals of Human Biology. 2003;30(5):570-578. doi:10.1080/0301446032000112670.

20. Chang CC, Chow CC, Tellier LC, Vattikuti S, Purcell SM, Lee JJ. Second-generation PLINK: rising to the challenge of larger and richer datasets. GigaScience. 2015;4:7. doi:10.1186/s13742-015-0047-8.

21. Pathak AK, Kadian A, Kushniarevich A, et al. The Genetic Ancestry of Modern Indus Valley Populations from Northwest India. American Journal of Human Genetics. 2018;103(6):918-929. doi:10.1016/j.ajhg.2018.10.022.

22. Moorjani P, Thangaraj K, Patterson N, et al. Genetic evidence for recent population mixture in India. American Journal of Human Genetics. 2013;93(3):422-438. doi:10.1016/j.ajhg.2013.07.006.

23. Reich D, Thangaraj K, Patterson N, Price AL, Singh L. Reconstructing Indian population history. Nature. 2009;461:489-494. doi:10.1038/nature08365.

Appendix A. Recommended reporting language

Avoid	Prefer
“The model is valid because p > 0.05.”	“The specified model was not rejected at alpha = 0.05 under this data and reference configuration.”
“The highest p-value is the true model.”	“Several models may be compatible; historical preference requires independent evidence.”
“AASI percentage” without qualification.	“AASI-related coefficient under the stated proxy/source configuration.”
“Iranian farmer ancestry” as a synonym for the whole Indus Periphery component.	“Indus Periphery-related ancestry,” or the exact source population used.
“Steppe ancestry proves Indo-European language.”	“Steppe_MLBA-related ancestry”; discuss language hypotheses separately.
Treating Vahaduo distance as a qpAdm p-value.	Report coordinate-space distance as a separate descriptive/exploratory measure.

Appendix B. Suggested run-manifest fields

run_id; analysis_phase (benchmark / sensitivity / exploratory); dataset_release; dataset_hashes; target; ordered_sources; right_populations; snp_panel; snp_handling_rule; engine; engine_version_or_commit; parameter_file_hash; input_snp_count; harmonized_snp_count; merged_snp_count; model_usable_snp_count; command; exit_code; stdout_path; stderr_path; runtime_seconds; coefficients; standard_errors; fit_p_value; status; notes.

Appendix C. Benchmark identifiers

ULL-KL = Ulladan, Kerala; PAL-TN = Palliyar, Tamil Nadu; IRU-TN = Irula, Tamil Nadu; LOH-GJ = Lohana, Gujarat; GUJA-GJ = GujaratiA study panel; YAD-RJ = Yadav_Rajasthan; BRH-HR = Brahmin_Haryana; RAJ-HR = Rajput; BHU-BR = Bhumihar_Bihar; YAD-UP = Yadav_UP; BHU-UP = Bhumihar_UP; BRH-UP = Brahmin_UP. These compact identifiers are manuscript cross-references, not official population labels.